VJTI Mumbai: Professional Chapters Guide

Professional chapters at VJTI provide students with global networking opportunities, access to international research journals, and platforms to participate in worldwide engineering competitions.

1. IEEE VJTI Student Branch (Institute of Electrical and Electronics Engineers)

The IEEE Student Branch at VJTI is one of the most active in the **IEEE Bombay Section**. It caters primarily to students from Electrical, Electronics, EXTC, Computer, and IT branches.

- **Core Focus:** Advancing technology for humanity through workshops on Python, Embedded Systems, Machine Learning, and VLSI.
- **Key Events:**
    - **Technical Paper Presentations (TPP):** Encouraging students to write and present research papers.
    - **Workshops:** Hands-on training on software like Eagle (PCB Designing) and MATLAB.
    - **IEEEXtreme:** A global 24-hour competitive programming challenge.
- **Member Benefits:** Access to the IEEE Xplore digital library, subsidized rates for international conferences, and global networking with industry experts.
- **Recruitment:** Usually held in **September**. It involves an aptitude test followed by a personal interview to join the Executive Committee.

2. ASME VJTI Student Section (American Society of Mechanical Engineers)

The ASME VJTI chapter is the hub for Mechanical and Production engineering students, focusing on professional development and multidisciplinary engineering.

- **Core Focus:** Mechanical design, energy systems, and robotics.
- **Key Events:**
    - **E-Fests (Engineering Festivals):** National and international level design competitions and hackathons.
    - **HPVC (Human Powered Vehicle Challenge):** Designing high-efficiency vehicles powered by human energy.
    - **SDC (Student Design Competition):** Solving real-world engineering problems through prototype building.
- **Member Benefits:** Access to *Mechanical Engineering* magazine, "A Day in the Life" webinar series, and eligibility for exclusive ASME scholarships.

- **Recruitment:** Membership is open year-round, but "Core Committee" selection happens in **August/September** through interviews and case-study rounds.

3. SAE VJTI Collegiate Chapter (Society of Automotive Engineers)

SAE VJTI is the governing body for all automotive-related technical teams in the college. It bridges the gap between classroom learning and the automotive industry.

- **Core Focus:** Vehicle dynamics, engine design, powertrain, and aerodynamics.
- **Key Events:**
    - **BAJA SAEIndia:** Designing and racing All-Terrain Vehicles (ATVs).
    - **SUPRA SAEIndia:** Designing Formula-style race cars.
    - **Effi-Cycle:** Developing energy-efficient hybrid tricycles.
- **Industry Interaction:** Regular seminars from companies like **Mahindra & Mahindra, Tata Motors, and L&T**.
- **Recruitment:** Focuses on technical knowledge of automobile components. The "Recruitment Drive" typically occurs in **late August** and includes a written test on basic mechanics and CAD concepts.

Comparative Overview of Professional Chapters

Chapter	Primary Department	Major Competition	Focus Area
IEEE	Electrical / IT / Electronics	IEEEXtreme / TPP	Electronics, AI & Software
ASME	Mechanical / Production	HPVC / SDC	Design & Core Engineering
SAE	Mechanical / Electrical	BAJA / SUPRA	Automotive & EV Tech